National Qualifications 2025

X835/77/11

Graphic Communication Supplementary sheets

WEDNESDAY, 30 APRIL

1:00 PM – 3:30 PM

Supplementary sheet 1 for use with question 1, supplementary sheets 2 and 3 for use with question 3 and supplementary sheet 4 for use with question 5.

SQA ©

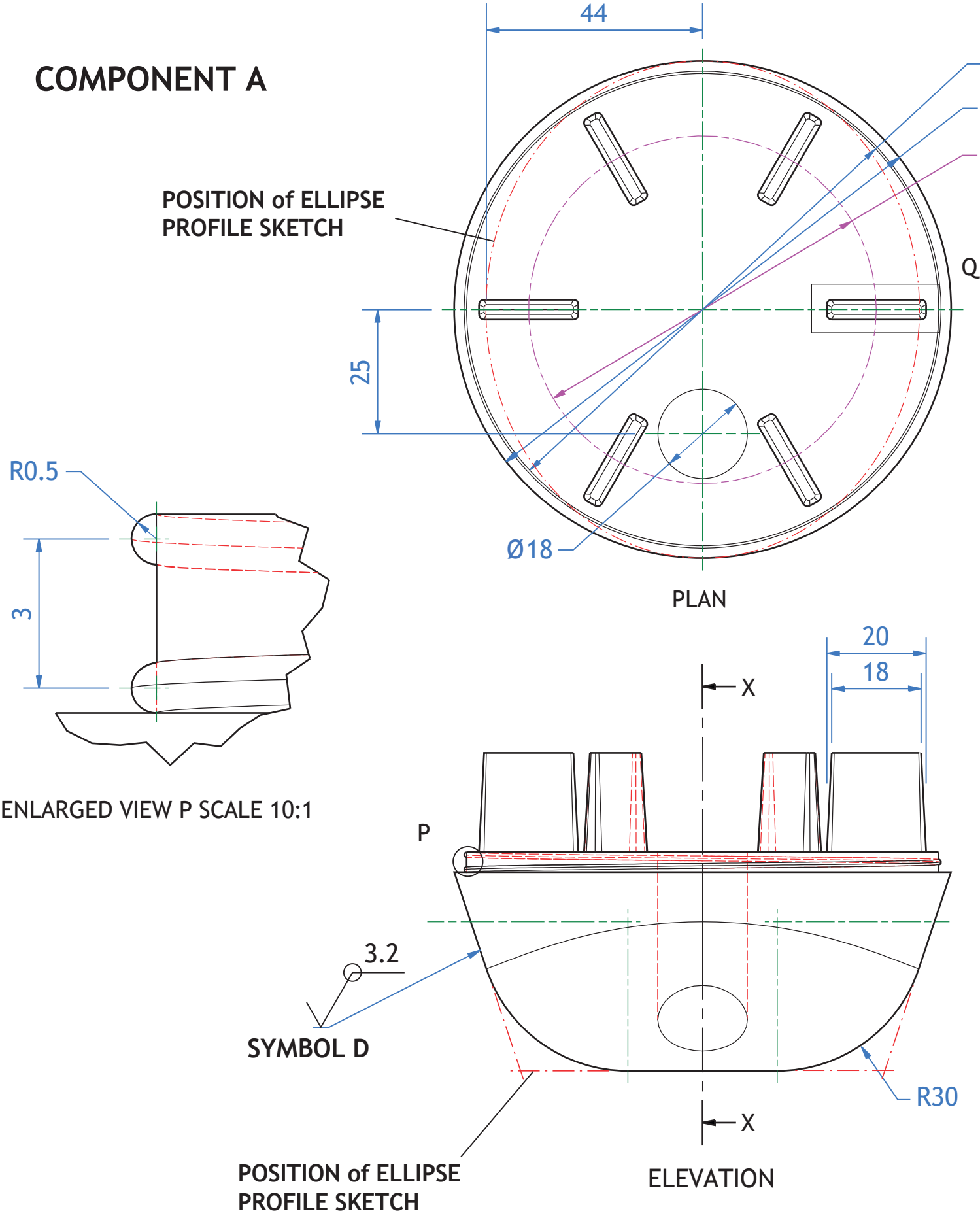
X8357711

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BIKE BUILDER CUSTOMISATION
ROAD BIKE
360°
3D INTERACTIVE VIEW
Small
5'5" - 5'7"
165 - 170cm
SELECT
Medium
5'7" - 5'10"
170 - 178cm
SELECT
Large
5'10" - 6'1"
178 - 185cm
SELECT
Frame Colour
Cyan d'Azur Flip
Voodoo Bronze Flip
Reflex Green Flip
Reflex Blue Flip

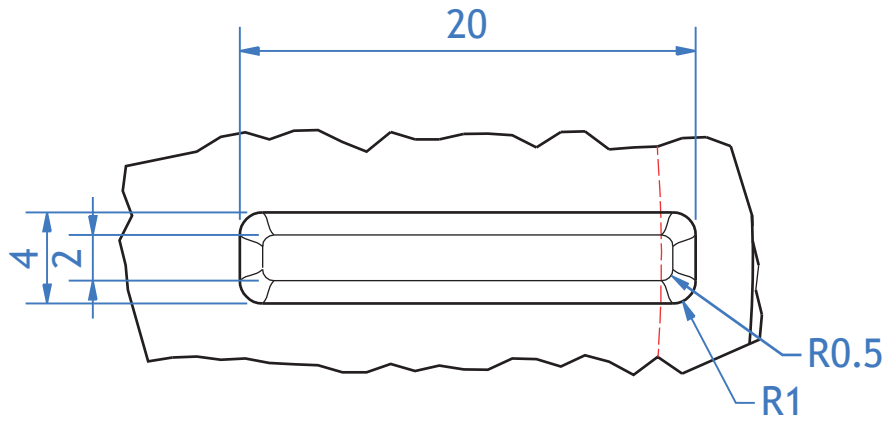
COMPONENT A


44
POSITION of ELLIPSE
PROFILE SKETCH
25
Ø18
PLAN
R0.5
3
ENLARGED VIEW P SCALE 10:1
20
18
X
P
3.2
SYMBOL D
R30
X
POSITION of ELLIPSE
PROFILE SKETCH
ELEVATION

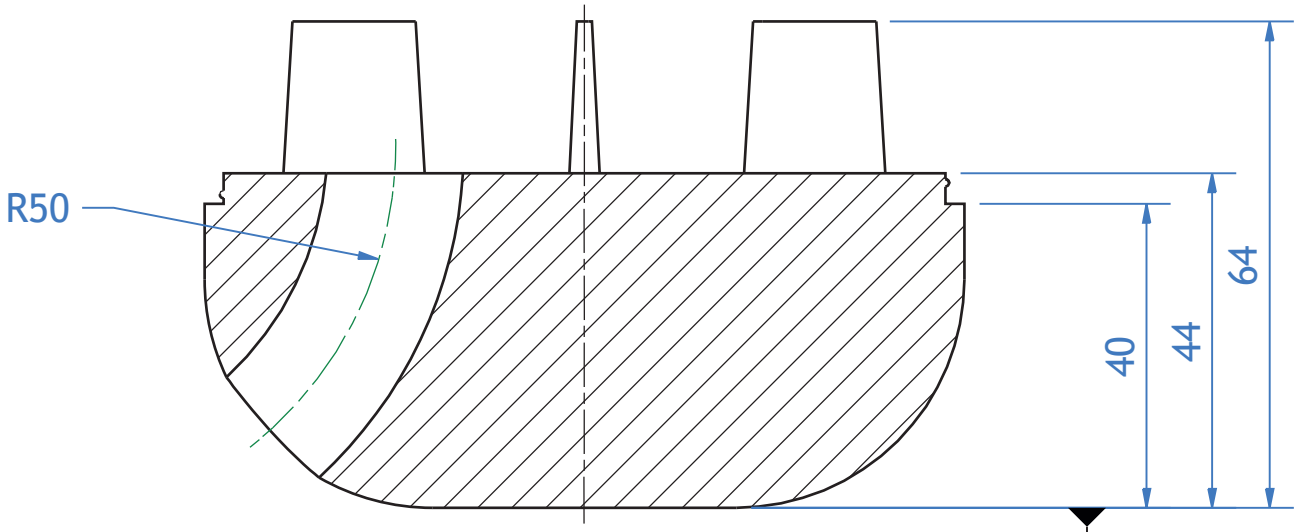
Ø95 ± 0.15 — TOLERANCE G

Ø100

PCD 70



ENLARGED VIEW Q SCALE 3:1 (CENTRE LINES REMOVED FOR CLARITY)

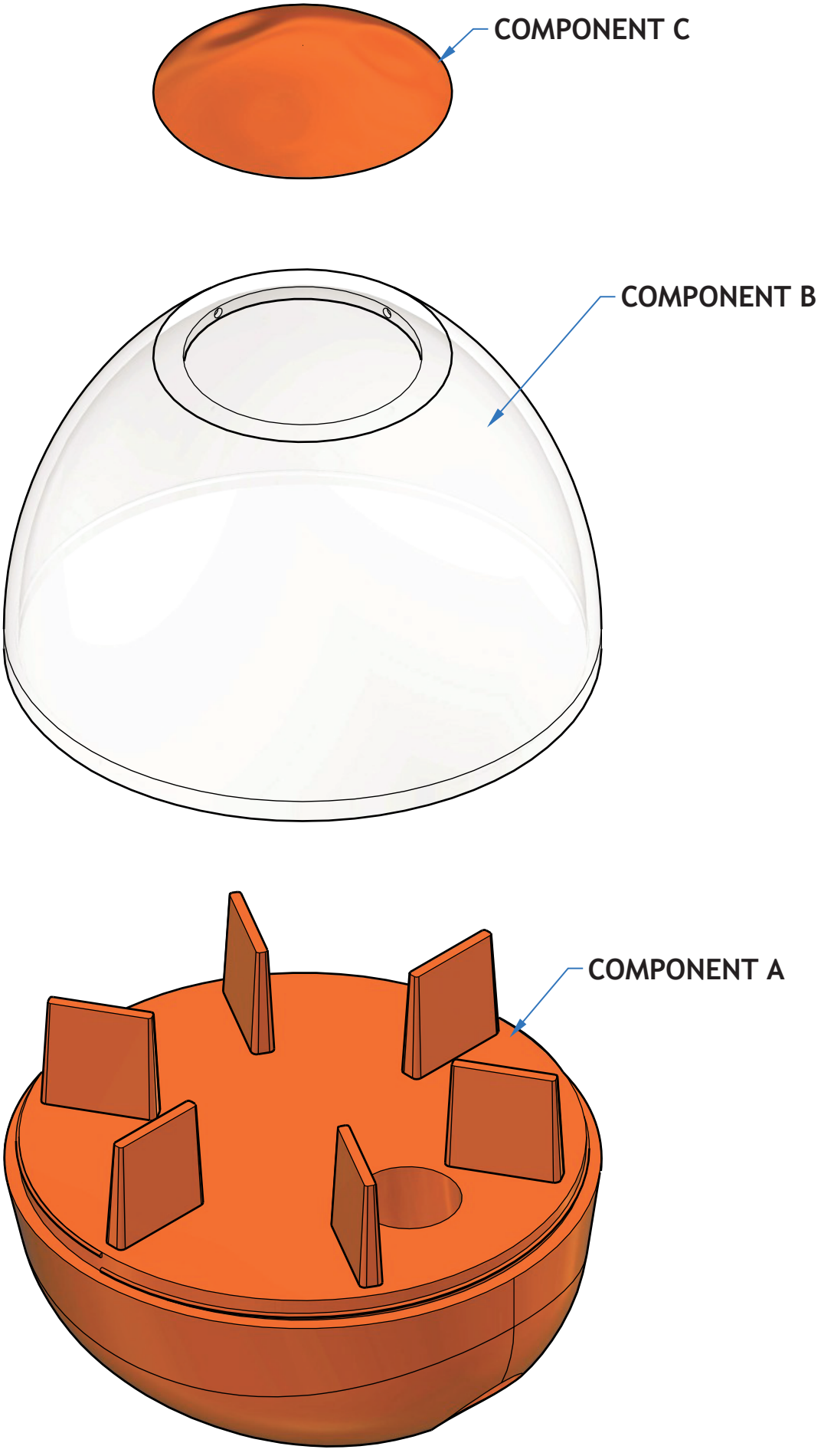


SECTION X-X

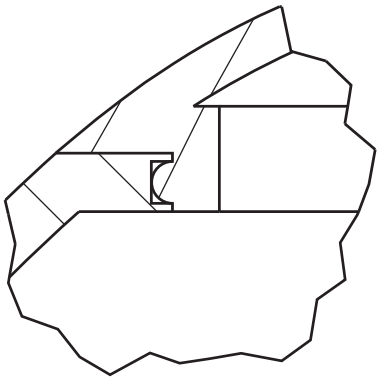
SYMBOL E

TOLERANCE F

TITLE: DOG TREAT TOY	
DRAWING NUMBER: 1	
ALL DIMENSIONS IN MM TOLERANCE ± 0.2 UNLESS OTHERWISE STATED	SCALE: 1:1
AUTHOR: A.NON	DATE: 10/10/23



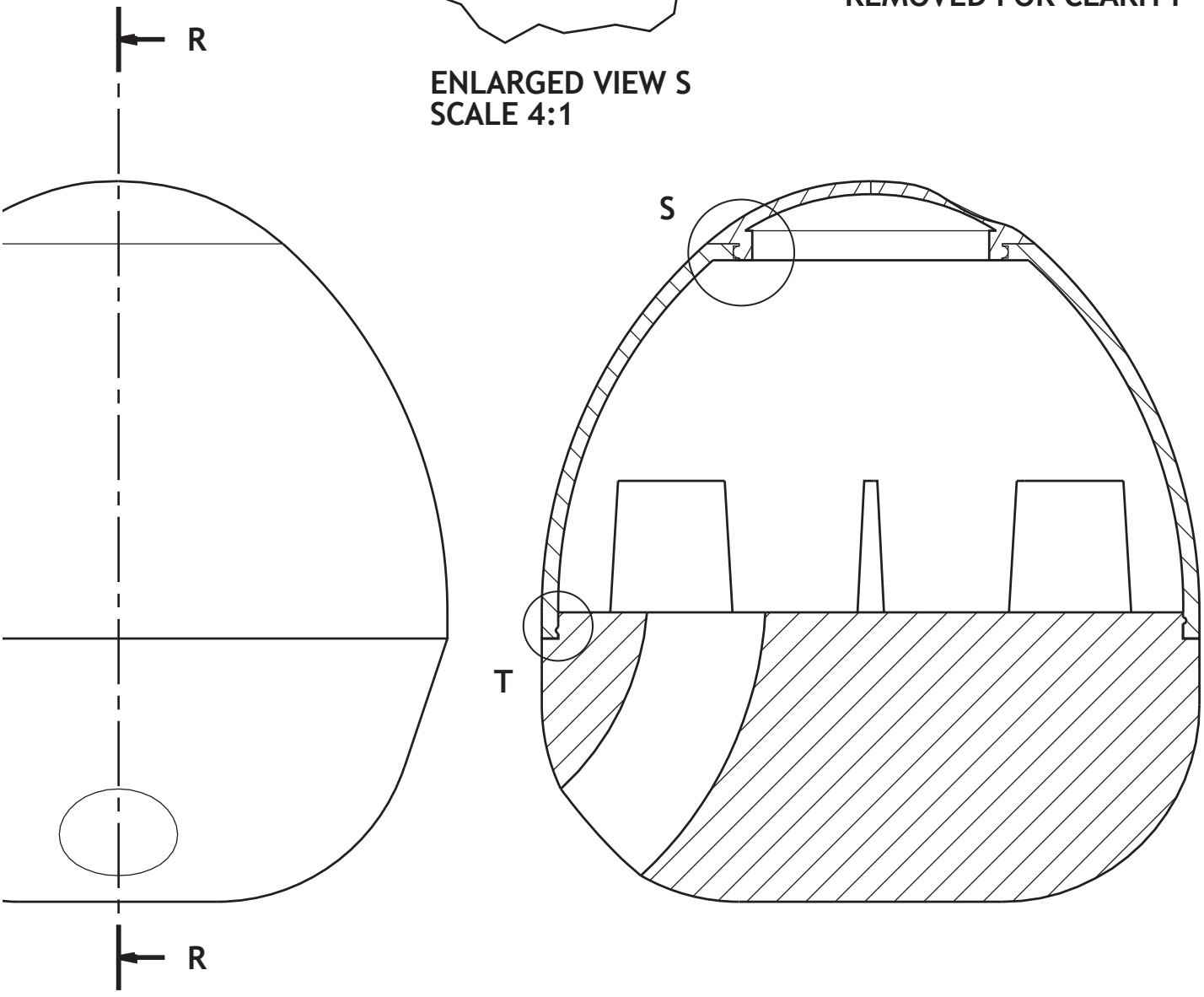
EXPLODED



ASSEMBLED

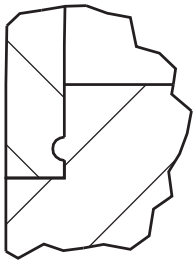
CENTRE LINES HAVE BEEN REMOVED FOR CLARITY

ENLARGED VIEW S
SCALE 4:1

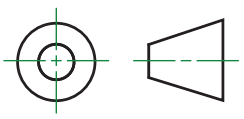


ELEVATION

SECTION R-R



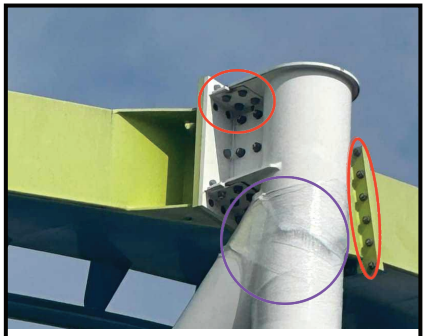
ENLARGED VIEW T
SCALE 4:1

TITLE: DOG TREAT TOY	
DRAWING NUMBER: 2	
EXPLODED AND ORTHOGRAPHIC VIEWS	SCALE: 1:1
AUTHOR: A.NON	DATE: 10/10/23

Specification

- Footprint: 66×40 m approx
- Height: 20.1 m
- First loop height: 20 m
- Second loop height: 15.4 m
- Track length: 425 m
- Max speed: 18 m/s (65 km/h)
- Acceleration max: 4.2 g
- Run time: 75/80 s
- Number of trains: 1 with 6 cars, 4 seats each
- Track material: Steel
- Foundation material: Concrete
- Foundation connected to supports using fixing bolts
- Frame, supports and tracks assembled using bolts and welding

Joining Methods

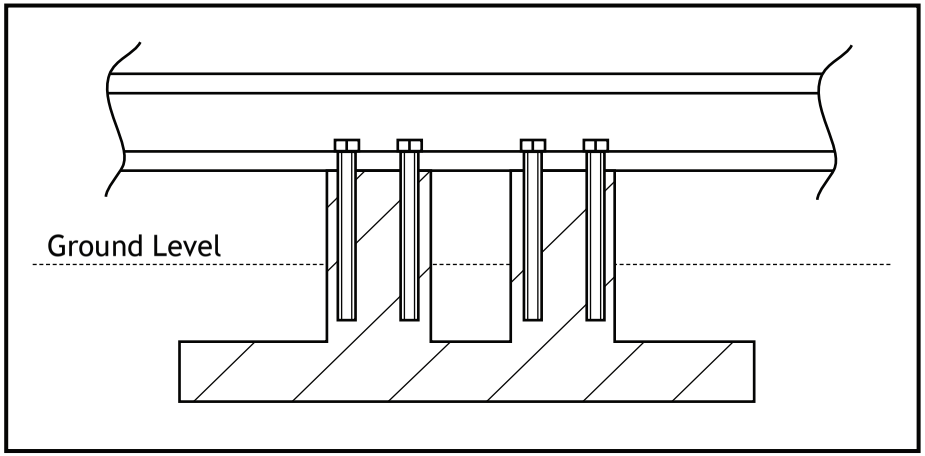


A: Fixing bolts with nut

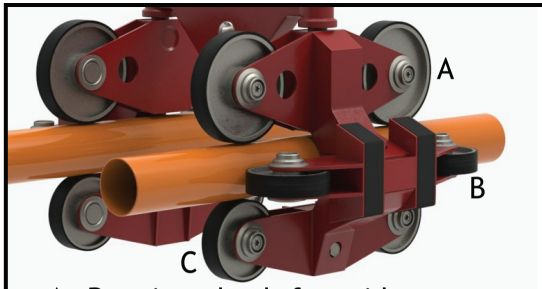
B: Welded steel supports

Column Foundation (Not to BS8888)





Mechanical wheel and brake system



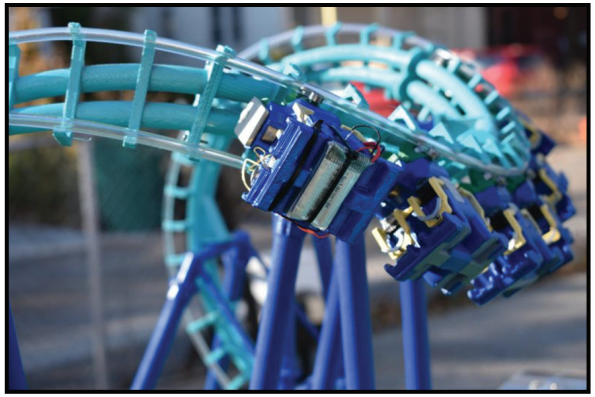
A: Running wheels for guidance

B: Side friction wheels for cornering

C: Up wheels to keep coaster on track

3D Printed model





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Supplementary sheet 1

Pay icon: https://pixabay.com/vectors/money-wallet-bank-banknote-bill-3722123/

Location icon: https://pixabay.com/vectors/location-position-job-you-are-here-1132648/

Test icon: https://pixabay.com/vectors/checkmark-tick-check-yes-mark-303752/

Home icon: https://pixabay.com/vectors/house-icon-home-symbol-sign-308936/

Bike helmet: https://pixabay.com/photos/bicycle-helmet-helmet-bike-helmet-505399/

Supplementary sheet 4

Joining methods: Image is taken from Davide Mavillonio on LinkedIn, https://www.linkedin.com/posts/davide-mavillonio_fem-cfd-ansys-ugcPost-7082302878328995840-Wn4t/

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Mechanical wheel and brake system: Image is taken from https://free3d.com/3d-model/roller-coasterwheel-884.html

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